On the Classification of Finite Groups with Maximal Subgroups of Prime Order

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Abstract

There are theorems for decomposing finite groups using their normal subgroups. The usefulness of these theorems reaches their limit at the introduction of simple groups (no normal subgroups to use). Simple groups can, contrary to what their name suggest, be extraordinarily complicated. The purpose of this article is to classify all members a potentially simpler type of group than simple groups. These are groups with a set number of subgroups, none being contained in any other (excluding the entire group and the trivial subgroup of course) which we call single layer groups. In the same way in which groups are constructed from simple groups, these groups are constructed from groups of prime order. The paper will show that no finite single layer groups are simple and that all of them either have the form of a cyclic group or semi-direct product of prime order groups.

1 Groups of Prime Order

Definition 1.1. *Let G be a group, any subgroup H such that $H \notin \{\{1\}, G\}$ is called a nontrivial subgroup of G, where $\{1\}$ denotes the trivial subgroup of G. The set of nontrivial subgroups of G will be denoted by $\mathcal{T}(G)$.*

Lemma 1.1. *Let G be a cyclic group of order n. For all $k \mid n$, $|g^k| = n/k$.*

Proof. Let G be a cyclic group of order n with generator g and let k divide n. Since $(g^k)^{n/k} = g^n = 1$, g^k must have order dividing n/k. If $d = |g^k|$ was less than n/k

then g would have order dividing dk and G would have order less than n which is a contradiction. $\square$

Theorem 1.1. *Let G be a group. There are no nontrivial subgroups of G if and only if $G \cong \mathbb{Z}/p\mathbb{Z}$ for some prime p or $G = \{1\}$.*

Proof. Let G be a group such that $\mathcal{T}(G) = \emptyset$. If $G = \{1\}$ then we're done, so assume $G \neq \{1\}$. There exists $g \in G$ such that $g \neq 1$ and so either $\langle g \rangle \in \mathcal{T}(G)$ or $\langle g \rangle = G$. Since $\mathcal{T}(G)$ is empty, $\langle g \rangle = G$. Suppose the order of G was composite. Let p be a prime such that $p \mid |G|$, we have that $\langle g^p \rangle$ has order $|G|/p$ which is less than $|G|$ and so $\langle g^p \rangle \in \mathcal{T}(G)$ or $\langle g^p \rangle = \{1\}$. Since $\mathcal{T}(G) = \emptyset$ it follows that $\langle g^p \rangle = \{1\}$ and so $g^p = 1$. This means that $|g| = |G|$ divides p but $p < |G|$ since $|G|$ is composite. This is a contradiction. Therefore, $|G|$ is prime.

Now, if $G = \{1\}$ then $\mathcal{T}(G) = \emptyset$ and we're done. Let G be a cyclic group of prime order and let $T \in \mathcal{T}(G)$. By Lagrange's Theorem $|T|$ must divide $|G|$ and so $|T| = 1$ or $|T| = |G|$. If $|T| = |G|$ then $T = G$, and if $|T| = 1$ then $T = \{1\}$. Both cases are contradictions and therefore $\mathcal{T}(G) = \emptyset$. $\square$

Corollary 1.1. *Any group of prime order is cyclic*

Proof. Let G be a group of prime order and suppose H is a nontrivial subgroup of G. We then have that $|H|$ divides $|G|$ by Lagrange's Theorem and so $|H| = |G|$ or $|H| = 1$. In either case H is a trivial subgroup of G and we've reached a contradiction. Therefore, no nontrivial subgroups of G exist and G must be cyclic. $\square$

Theorem 1.2. *Let G be a group and let $c \in G$. If $c \neq 1$ and c is not contained in any $T \in \mathcal{T}(G)$. Then G is generated by c.*

Proof. Let G be a group and let $c \in G$ with $c \neq 1$. Suppose that c is not contained in any $T \in \mathcal{T}(G)$. We then have that $\langle c \rangle \neq \{1\}$ and $\langle c \rangle \notin \mathcal{T}(G)$. This means that $\langle c \rangle = G$ and we're done. $\square$

Corollary 1.2. *If G is a group of prime order then all non-identity elements of G generate G*

2 Actions

Definition 2.1. *Given a set X we define* $\mathrm{Bij}(X)$ *to be the group of bijections from X to itself.*

Theorem 2.1. *Let X be a finite set and let p be a prime such that p does not divide $|X|$. For any bijection $\phi : X \to X$ such that ϕ has order p in* $\mathrm{Bij}(X)$*, ϕ has a fixed point.*

Proof. Let X be a set and induct on the cardinality of X. If $|X| < p$ then $|\mathrm{Bij}(X)| = |X|!$ has no factor of p and so, by Lagrange's Theorem, there can be no element $\phi \in \mathrm{Bij}(X)$ of order p. Due to this, the base case is vacuously satisfied. Suppose there exists $N > p$ indivisible by p such that if $|X| < N$ and p does not divide $|X|$ then any element ϕ of order p in $\mathrm{Bij}(X)$ has a fixed point. Let $|X| = N$ and suppose ϕ is an element of order p in $\mathrm{Bij}(X)$. Pick $x \in X$ and suppose that $\phi^n(x) = x$ for some $0 < n < p$. We'd then have that $\phi^n \in \{b \in \mathrm{Bij}(X) : bx = x\}$. The set $\{b \in \mathrm{Bij}(X) : bx = x\}$ is a subgroup of $\mathrm{Bij}(X)$ and so is $\langle\phi\rangle \cap \{b \in \mathrm{Bij}(x) : bx = x\}$ which also contains ϕ^n. Since ϕ has order p, $\langle\phi\rangle$ also has order p and $\mathcal{T}(\langle\phi\rangle) = \emptyset$. Therefore, either $\langle\phi\rangle = \langle\phi\rangle \cap \{b \in \mathrm{Bij}(X) : bx = x\}$ or $\langle\phi\rangle \cap \{b \in \mathrm{Bij}(X) : bx = x\} = \{1\}$ which means that $\phi(x) = x$ or $\phi^n(x) \neq x$ for all $0 < n < p$. Our assumptions contradict the idea that $\phi^n(x) \neq x$ for all $n < p$ and so we must conclude that $\phi(x) = x$ which shows that ϕ has x as a fixed point.

Now suppose that $\phi^n(x) \neq x$ for all $0 < n < p$ and let $0 < n \leq m \leq p$. If $\phi^n(x) = \phi^m(x)$ then since ϕ is a bijection it follows that $\phi^{m-n}(x) = x$. This means that $m - n = 0$ since $0 \leq m - n < p$. Hence, $m = n$. Therefore, the set $O_x = \{\phi^n(x) : 0 < n \leq p\}$ has exactly p elements. Let $X' := X - O_x$. Since $|X| = N > p$, we have that $X' \neq \emptyset$. Suppose there exists $y \in X'$ such that $\phi(y) \in O_x$. We then have that $\phi(y) = \phi^n(x)$ for some $n \leq p$ and so $\phi^{p-1}(\phi(y)) = y = \phi^{p+n-1}(x)$. If $n = 1$ then $y = x$, otherwise, $p > n - 1 > 0$ and so $y = \phi^{n-1}(x) \in O_x$ which is a contradiction. Conversely, suppose there exists $y \in X'$ such that $\phi(z) = y$ for some $z \in O_x$. Then $\phi^{n+1}(x) = y$ for some $n \leq p$. If $n < p$ then $y \in O_x$ and if

$n = p$ then $\phi(x) = y$ and $y \in O_x$. Therefore, $\phi' : X' \to X'$ such that $\phi'(y) = \phi(y)$ for all $y \in X'$ is a bijection. Suppose ϕ' did not have order p in $\mathrm{Bij}(X')$. Since for all $y \in X'$, $\phi^p(y) = y$, it follows that ϕ' must at least have order dividing p. This means that the order of ϕ' must be either 1 or p. If the order of ϕ' is 1 then ϕ' is the identity on X' and every point in X' is fixed by ϕ'. If the order of ϕ' is p then since $|X'| = |X| - |O_x| = |X| - p$, $|X'|$ is still not divisible by p and we may use the inductive hypothesis to conclude that ϕ' must have a fixed point in X'. Since ϕ' has a fixed point in X', ϕ must have a fixed point in X and we're done. □

3 Single Layer Groups

Definition 3.1. *A group G is a single layer group if $\mathcal{T}(G) \neq \emptyset$ and for all $A, B \in \mathcal{T}(G)$, B does not properly contain A. In this case, the elements of $\mathcal{T}(G)$ are called the parallel subgroups of G.*

Theorem 3.1. *Let G be a group. The following are equivalent.*

1. *G is a single layer group*
2. *$\mathcal{T}(G)$ is nonempty and every element $T \in \mathcal{T}(G)$ is minimal under set inclusion.*
3. *$\mathcal{T}(G)$ is nonempty and every element $T \in \mathcal{T}(G)$ is maximal under set inclusion.*
4. *There exists non-identity elements $a, b \in G$ such that $a \notin \langle b \rangle$ and for any such pair $\langle a, b \rangle = G$.*
5. *$\mathcal{T}(G)$ is nonempty and each $T \in \mathcal{T}(G)$ has prime order.*

Proof. Let G be a single layer group. We have that $\mathcal{T}(G)$ is nonempty. Let $T \in \mathcal{T}(G)$. There does not exist $A \in \mathcal{T}(G)$ which is properly contained by T and so T is minimal under set inclusion.

Let G be a group such that $\mathcal{T}(G)$ is nonempty and for all $T \in \mathcal{T}(G)$, T is minimal in $\mathcal{T}(G)$ under set inclusion. Let $T \in \mathcal{T}(G)$ and suppose there exists $H \in \mathcal{T}(G)$ such that H properly contains T. We then have that H properly contains an element

of $\mathcal{T}(G)$ and so H is not minimal in $\mathcal{T}(G)$ under set inclusion. We then conclude that there is no $H \in \mathcal{T}(G)$ such that H properly contains T and so T is maximal in $\mathcal{T}(G)$ under set inclusion.

Now suppose that G is a group such that $\mathcal{T}(G)$ is nonempty and all elements of $\mathcal{T}(G)$ are maximal under set inclusion. Since $\mathcal{T}(G)$ is nonempty we have that there exists $H \neq \{1\}$ a subgroup of G which does not contain some element $g \in G$. This means that for any $h \in H$, $g \notin \langle h \rangle$. This shows the existence part of claim four. For the second part of claim four, pick $a, b \in G$ such that $a \notin \langle b \rangle$, then $\langle b \rangle \in \mathcal{T}(G)$. Since $b \in \langle a, b \rangle$, either $\langle a, b \rangle \in \mathcal{T}(G)$ or $\langle a, b \rangle = G$. If $\langle a, b \rangle \in \mathcal{T}(G)$ then $\langle b \rangle$ is properly contained in an element of $\mathcal{T}(G)$ which means that $\langle b \rangle$ is not maximal, which is a contradiction. Thus $\langle a, b \rangle = G$.

Now, let G be a group and suppose that there exists non-identity elements $a, b \in G$ such that $a \notin \langle b \rangle$ and for any such pair, $\langle a, b \rangle = G$. Let $T \in \mathcal{T}(G)$. If T is not cyclic then for any non-identity $t \in T$ there exists $t' \in T$ such that $t' \notin \langle t \rangle$. However, this means that $G = \langle t, t' \rangle \subseteq T$ which contradicts T being nontrivial. Therefore, T is cyclic. If T was infinite then given a generator t of T, $t^2 \notin \langle t^4 \rangle$ and so $\langle t^2, t^4 \rangle = G$. This means that $T = G$ which is again a contradiction. We then have that T is finite. If $|T|$ is composite then by Cauchy's Theorem, given a prime p which divides $|T|$, there exists a subgroup H of T of order p. Since the order of H does not equal the order of T, H must be a proper subgroup and so there exists $t \in T$ such that $t \notin H$. Let h generate H. This means that $G = \langle h, t \rangle \subseteq T$ which is a contradiction. Therefore, $|T|$ must be prime.

Finally, suppose that G is a group such that $\mathcal{T}(G)$ is nonempty and each $T \in \mathcal{T}(G)$ has prime order. Let $A, B \in \mathcal{T}$ such that $A \subseteq B$. Since B has prime order, $A = B$ and so B does not properly contain A. This is sufficient to show that G is a single layer group. □

Theorem 3.2. *Let G be a finite single layer group. Either the order of G is a product*

of primes with multiplicity one for each prime factor, or all parallel subgroups of G have the same order.

Proof. Let G be a finite single layer group. Suppose there exists a prime p such that $p^k \mid |G|$ with $k > 1$. Let k be maximal such that $p^k \mid |G|$. If $|G| = p^k$ then we're done as all intermediate subgroups must have order p. If not, then there exists d coprime to p such that $|G| = dp^k$ for some $k > 1$ and by the Sylow Theorems, there exists a proper subgroup H of G with order p^k. Since p^k is not prime ($k > 1$) we have by Cauchy's Theorem that H contains a proper subgroup H' of order p. Therefore, H properly contains another parallel subgroup which is a contradiction. Hence, if $p \mid |G|$ then p must have multiplicity one in $|G|$. □

Theorem 3.3. *Let G be a group. $\mathcal{T}(G)$ is a singleton if and only if $G \cong \mathbb{Z}/p^2\mathbb{Z}$ for some prime p*

Proof. Let G be a group. Suppose $G \cong \mathbb{Z}/p^2\mathbb{Z}$ for some prime p. Since p^2 is not prime, G has at least one nontrivial subgroup. Let $H \in \mathcal{T}(G)$ and let $h \in H$ such that $h \neq 1$. We have that H has order dividing p^2. The order of H is not p^2 nor 1 so the order of H must be p. Let g generate G. We then have that $h = g^k$ for some $k < p^2$. We then have that $g^{kp} = 1$ and so $kp \mid p^2$. This means that $k \mid p$ and so $k = 1$ or $k = p$. If $k = 1$ then $h = g$ and so $H = G$ which is a contradiction. Therefore, $h = g^p$. Since all non-identity elements of a group of prime order are generators, we have that $H = \langle g^p \rangle$. Conversely, let G be a group such that $\mathcal{T}(G)$ is a singleton. We then have that for all $g \in G$ either $g \in T$ where T is the element of $\mathcal{T}(G)$ or g generates G. There must be some element $g \notin T$ since T is a proper subgroup of G and so we may conclude that G is cyclic with generator g. If G is an infinite cyclic group, then G has more than one element of $\mathcal{T}(G)$ which is a contradiction (this was shown in the proof of Theorem 3.1). Therefore, G must be a finite group and by Cauchy's Theorem, must have subgroups of order q for every prime q which divides G. It then follows that there must be exactly one prime p which divides G. Suppose the order of G was p^k for some $k > 2$. The order of $g^{p^{k-2}}$ is p^2 and the order of $g^{p^{k-1}}$ is p. This means that $\langle g^{p^{k-2}} \rangle$ and $\langle g^{p^{k-1}} \rangle$ are distinct

elements of $\mathcal{T}(G)$ which is a contradiction. Therefore, $k \leq 2$. If $k = 1$ then G is a group of order p and $\mathcal{T}(G) = \emptyset$, which is also a contradiction. We then conclude that $k = 2$. □

Theorem 3.4. *Let G be a group. G is a non-simple single layer group if and only if $G \cong \mathbb{Z}/p^2\mathbb{Z}$ or $G \cong \mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ with p, q primes.*

Proof. Let G be a group. If $G \cong \mathbb{Z}/p^2\mathbb{Z}$ then G is not simple and is a single layer group. Suppose that $G \cong \mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ with p, q primes. Then $|G| = pq$ and so, due to Lagrange's Theorem, every element T of $\mathcal{T}(G)$ must have prime order and G is a single layer group. Since $\mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ has a nontrivial normal subgroup, G is not simple.

Conversely, let G be a non-simple single layer group. There exists a normal subgroup $T \in \mathcal{T}(G)$ of G. Since T is maximal in $\mathcal{T}(G)$ the only subgroups of G containing T are G and T itself. Because of the correspondence between subgroups of G/T and subgroups of G containing T, we can conclude that $\mathcal{T}(G/T) = \emptyset$ and so G/T must be trivial or have prime order. In either case, there are a finite number of cosets of T in G and since T is a finite group, this means that G must also be finite. We then have that $\mathcal{T}(G)$ has a well defined cardinality.

If $|\mathcal{T}(G)| < 2$ then $G \cong \mathbb{Z}/p^2\mathbb{Z}$ for some prime p. Suppose $|\mathcal{T}(G)| \geq 2$, we then have that there exists $A, T \in \mathcal{T}(G)$ such that $A \neq T$ and T is normal in G. Let $\pi : A \times T \to G$ be the set map such that $\pi(a, t) = at$ for all $a \in A$ and $t \in T$. Endow $A \times T$ with the group operation $\star$ such that $(a, t) \star (a', t') = (aa', (a')^{-1}ta't')$. Observe that

$$\pi((a, t) \star (a', t')) = aa'(a')^{-1}ta't' = ata't' = \pi(a, t)\pi(a', t')$$

This means that π is a homomorphism from $A \times T$ endowed with the product $\star$ to

G. Take $a \in A$ and $t \in T$. $\pi(a,1) = a$ and $\pi(1,t) = t$. This means that

$$G = \langle A \cup T \rangle \subseteq \pi(A \times T)$$

where $G = A \cup T$ since A and T are distinct. This means that π is surjective. Let $(a,t) \in \ker \pi$, we then have that $at = 1$ and $a = t^{-1}$. Since A and T have no nontrivial subgroups we know that $A \cap T = \{1\}$ and this means that $a = 1 = t$. Hence, the kernal of π is trivial and π is an isomorphism. Astute readers will recognize that $A \times T$ with the operation $\star$ is precisely the (inner) semidirect product $A \ltimes T$ (where T is the normal subgroup). Since both A and T have prime order $A \ltimes T \cong \mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ where p and q are primes. □

Corollary 3.1. *Let G be a single layer group with a nontrivial normal subgroup. Then $|G| = pq$ where p and q are primes.*

Theorem 3.5. *Let G be a group. G is a single layer group with $|\mathcal{T}(G)| = 2$ if and only if $G \cong \mathbb{Z}/pq\mathbb{Z}$ for two distinct primes p and q.*

Proof. Let G be a single layer group such that $|\mathcal{T}(G)| = 2$. Take A and B to be the two distinct elements of $\mathcal{T}(G)$ and let $a \in A$ and $b \in B$ be non-identity elements of each. It follows immediately from the definition of a single layer group that A and B must intersect trivially. If $ab = 1$ then it follows by closure that $b \in A$ and $a \in B$. Since $A \cap B$ is trivial $a = 1 = b$ which is a contradiction. Therefore, $ab \neq 1$. If $ab \in A$ then $b \in A$ and similarly, if $ab \in B$ then $a \in B$. Both these statements contradict the fact that A and B intersect trivially and so $ab \notin T$ for any $T \in \mathcal{T}(G)$. This means that ab must generate G. Since G is cyclic and $\mathcal{T}(G)$ is nonempty, G cannot be simple. This means that the order of G is pq for some primes p and q. If $p = q$ then $G \cong \mathbb{Z}/p^2\mathbb{Z}$ and $|\mathcal{T}(G)| = 1$ which contradicts our assumption. Thus $p \neq q$ and $G \cong \mathbb{Z}/pq\mathbb{Z}$.

Conversely, let $G \cong \mathbb{Z}/pq\mathbb{Z}$. We have that for all $T \in \mathcal{T}(G)$, $|T|$ must divide pq. Since $T \in \mathcal{T}(G)$, $|T| \neq 1$ and $|T| \neq pq$ and so it follows that $|T| = p$ or $|T| = q$. Let g generate G. Since $\langle g^p \rangle$ is a nontrivial subgroup of G, $\mathcal{T}(G)$ is nonempty and

each element $T \in \mathcal{T}(G)$ has prime order. We then conclude that G is a single layer group. Since p and q have multiplicity one in the order of G, all groups of order p and all groups of order q are Sylow p-subgroups and Sylow q-subgroups respectively. Because G is abelian, there must be exactly one Sylow p-subgroup and one Sylow q-subgroup. It then follows that G must have exactly one group of order p and one group of order q. Any element $T \in \mathcal{T}(G)$ must have order p or q and therefore must be one of the two Sylow subgroups of G. This means that $\mathcal{T}(G)$ must have exactly two elements, and we're done. □

Theorem 3.6. *Let G be a single layer group of order p^k for some $k \in \mathbb{N}$. We have that $k = 2$ and $G \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ or $\mathbb{Z}/p^2\mathbb{Z}$.*

Proof. Let G be a single layer group of order p^k for some $k \in \mathbb{N}$. Since $\mathcal{T}(G)$ is nonempty, $k > 1$. Let $T \in \mathcal{T}(G)$. We have that $|T| = p$ since the order of T must be prime and p is the only prime dividing G. Suppose there exists $t \in T$ such that there is some coset $gT \in G/T$ with $gT \neq T$ such that $tgT = T$. We then have that $tg \in T$ and then $g \in T$. This contradicts the assumption that $gT \neq T$ and so no such t exists. It follows that T fixes the set $G/T - \{T\}$ under translation on cosets. Let $\Pi = G/T - \{T\}$ and let $\pi : T \to \mathrm{Bij}(\Pi)$ such that $\pi(t)(gT) = tgT$ for all $t \in T$ and $gT \in \Pi$. Since translation by t is a bijection on G/T which fixes Π for all $t \in T$, $\pi(t)$ is indeed a bijection from Π to itself. We also have that π is a homomorphism of groups since

$$\pi(tt')(gT) = tt'gT = \pi(t) \circ \pi(t')(gT)$$

for all $t, t' \in T$ and all $gT \in \Pi$. Let $t \in T$ with $t \neq 1$, it follows that the order of $\pi(t)$ is either p or 1. If $|\pi(t)| = 1$ then $\pi(t) = \mathrm{id}_\Pi$ and so

$$tgT = gT \text{ and } a^{-1}ta \in T$$

for all $gT \in \Pi$ and $a \in gT$. Since $t \neq 1$ and T has prime order, t generates T. Now, observe that given $a \in G$, if $a^{-1}ta \in T$ then, due to the closure of T, $a^{-1}t^k a \in T$ for all k and hence, $a^{-1}Ta \subseteq T$. Since T is finite and conjugation by any element

$a \in G$ is a bijection from T to $a^{-1}Ta$, it follows that $a^{-1}Ta = T$ when $a^{-1}ta \in T$. This means that for all $gT \in \Pi$ and $a \in gT$

$$a^{-1}Ta = T$$

and so T must be fixed by all elements $a \in G - \{T\}$ under conjugation. Since T also fixes T under conjugation, we may conclude that T is normal in G. Now, suppose that $|\pi(t)| = p$. Observe that

$$|\Pi| = |G/T - \{T\}| = |G/T| - 1 = p^{k-1} - 1$$

Since $k > 1$, $|\Pi| \neq 0$ and so we may conclude that p does not divide $|\Pi|$. Since $|\pi(t)|$ has order p in $\mathrm{Bij}(\Pi)$, $\pi(t)$ must have a fixed point in Π. This means that there exists $gT \in \Pi$ such that

$$tgT = gT \text{ and } a^{-1}Ta = T$$

for all $a \in gT$. Let $a \in gT$. Since $gT \neq T$, $a \neq 1$. Notice that since $a \notin T$ and T is generated by t, $a \notin \langle t \rangle$ and so $\langle a, t \rangle = G$. Since a fixes T under conjugation and t fixes T under conjugation, both a and t must be in the normalizer $N(T)$ of T. It follows that $\langle a, t \rangle = G \subseteq N(T)$ and so T is normal in G.

We have just shown that G has a normal subgroup. We then have that the order of G must be the product of two primes and so $k = 2$. If G is cyclic then $G \cong \mathbb{Z}/p^2\mathbb{Z}$ and we're done. If not, then observe that $|G|$ is not a product of distinct primes and so $|\mathcal{T}(G)| > 2$. We then have that there must exist at least two groups of order p in $\mathcal{T}(G)$ and that one of them is normal. Let $A, T \in \mathcal{T}(G)$ with T normal in G and $A \neq T$. We then have that for all $a \in A$, the map $\phi(a) : T \to T$ such that $\phi(a) = ata^{-1}$ for all $t \in T$ is an automorphism of T. Let $a \in A$ with $a \neq 1$. Recall that the automorphism group of a group of order p is a cyclic group of order $p - 1$. This means that the order of $\phi(a)$ must divide $p - 1$. Since $\phi(a)^p(t) = a^p t a^{-p} = t$ for any $t \in T$ we have that the order of $\phi(a)$ must divide p. Because $p - 1$ and p are coprime, the order of $\phi(a)$ in $\mathrm{Aut}(T)$ must be 1 and $\phi(a)$ must act trivally on T

under conjugation. In other words, a must commute with all elements of T. This means that $a \in C(T)$ where $C(T)$ is the centralizer of T. Since T is a group of prime order, it is cyclic and abelian. Therefore, $t \in C(T)$. Since $A \cap T = \{1\}$ and $a \neq 1$, we have that $a \notin \langle t \rangle$ and so $G = \langle t, a \rangle \subseteq C(T)$ and $G = C(T)$. This means that T is in the center of G.

Let $\Psi : A \times T \to G$ such that $\Psi(a, t) = at$ for all $a \in A$ and $t \in T$. Since T is in the center of G, T and A commute, and we have that

$$\Psi((a,t)(a',t')) = aa'tt' = ata't' = \Psi(a,t)\Psi(a',t')$$

and so Ψ is a homomorphism of groups. It is clear that $A \cup T \subseteq \Psi(A \times T)$ (consider the image of $(a, 1)$ or $(1, t)$ under Ψ) and so $\Psi(A \times T) = G$ due to G being a single layer group and A and T being distinct. This means that Ψ is surjective. Suppose there exists $(a, t) \in \ker \Psi$, then $at = 1$ and so $a = t^{-1}$. If $a \neq 1$ then $a \in A \cap T$ which contradicts the fact that A and T are distinct. Therefore, $a = t = 1$ and $\ker \Psi$ is trivial. This proves that Ψ is an isomorphism from $A \times T \to G$ and so $G \cong A \times T \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ and we're done. □

Theorem 3.7. *Let G be a group such that $G \cong \mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ where p and q are primes with $p > q$. The set $\mathcal{T}(G)$ has either 2 or $p+1$ elements and if $|\mathcal{T}(G)| = p+1$ then there is one group of order p and p groups of order q in $\mathcal{T}(G)$.*

Proof. Let p and q be primes with $p > q$. Let G be the group $\mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ with the left factor being the normal subgroup. Since p and q are distinct, all subgroups of prime order in G are Sylow subgroups of G and vice versa. This means that all subgroups of order p must be conjugate and all subgroups of order q must be conjugate. We then have that $\mathbb{Z}/p\mathbb{Z} \times \{1\}$ is the only subgroup of order p since it is normal in G. By closure $\{1\} \times \mathbb{Z}/q\mathbb{Z}$ fixes itself under conjugation. Let $N_p = \mathbb{Z}/p\mathbb{Z} \times \{1\}$ and let $N_q = \{1\} \times \mathbb{Z}/p\mathbb{Z}$. We have that $G = N_p N_q$ and so for any $g \in G$,

$$gN_qg^{-1} = nmN_qm^{-1}n^{-1} = nN_qn^{-1}$$

where $n \in N_p$ and $m \in N_q$ such that $g = nm$. We then have that $\{gN_qg^{-1} : g \in G\} = \{nN_qn^{-1} : n \in N_p\}$ and so $\{nN_qn^{-1} : n \in N_p\}$ is the set of all subgroups of G with order q. Since $|N_p| = p$ we have that there are at most p subgroups of order q. If there are exactly p subgroups of order q in G then we are done. If there are less than p subgroups of order q in G then for each $n \in N_p$ there exists $0 < k < p$ such that $n^k N_q n^{-k} = N_q$. We then have that $N_p \subseteq N(N_q)$ where $N(N_q)$ is the normalizer of N_q since some generator of N_p fixes N_q under conjugation. It follows that $N(N_q) = G$ since N_q normalizes itself and $N_pN_q = G$. This means that N_q and N_p are normal subgroups of G such that $N_pN_q = G$ and $N_p \cap N_q = \{1\}$ (N_p and N_q both have prime order), therefore $G \cong N_p \times N_q$. Since $p \neq q$ we have by the Chinese Remainder Theorem that $N_p \times N_q \cong \mathbb{Z}/pq\mathbb{Z}$ and so $|\mathcal{T}(G)| = 2$. □

Theorem 3.8. *Let G be a finite single layer group with $|\mathcal{T}(G)| > 2$. The sets $T - \{1\}$ with $T \in \mathcal{T}(G)$ partition $G - \{1\}$ and*

$$|G| = 1 + \sum_{T \in T(G)} (|T| - 1)$$

Proof. Let G be a finite single layer group with $|\mathcal{T}(G)| > 2$. Let $A, B \in \mathcal{T}(G)$. Either $A \cap B = A = B$ or $A \cap B = \{1\}$ since both A and B have prime order. We then have that $A - \{1\}$ and $B - \{1\}$ are disjoint for any pair $A, B \in \mathcal{T}(G)$ with $A \neq B$. If there exists $c \in G$ such that c is not in any parallel subgroup of G then c generates G and G is cyclic. Since $\mathcal{T}(G)$ is nonempty and G is abelian, $\mathcal{T}(G)$ contains a normal subgroup of G. We then conclude that $|G|$ is the product of two primes. It then follows that $|\mathcal{T}(G)| \in \{1, 2\}$ which is a contradiction. Therefore, for all $g \in G$ with $g \neq 1$ there exists $T \in G$ such that $g \in T$. This means that G is partitioned by the set $\{T - \{1\} : T \in \mathcal{T}(G)\}$. We then have that

$$|G - \{1\}| = |G| - 1 = \sum_{T \in \mathcal{T}(G)} |T - \{1\}| = \sum_{T \in \mathcal{T}(G)} (|T| - 1)$$

and

$$|G| = 1 + \sum_{T \in \mathcal{T}(G)} (|T| - 1)$$

□

Lemma 3.1. *Let G be a finite simple single layer group with $T \in \mathcal{T}(G)$, then*

$$\{gTg^{-1} : g \in G\} \cong G/T$$

Proof. Let G be a simple finite single layer group and let $T \in \mathcal{T}(G)$. Consider $\pi : \{gTg^{-1} : g \in G\} \to G/T$ such that $\pi(gTg^{-1}) = gT$ for all $g \in G$. We first check that this map is well defined. Let $g, h \in G$ such that $gTg^{-1} = hTh^{-1}$. We then have that $h^{-1}g \in N(T)$ where $N(T)$ is the normalizer of T. Since G is simple $N(T)$ must be a proper subgroup of G. We may then conclude that $N(T) = T$ by the maximality of $T \in \mathcal{T}(G)$. This means that $h^{-1}g \in T$ which implies that $gT = hT$ and so the map π is well defined. Suppose $gT = g'T$ with $gTg^{-1} \neq g'T(g')^{-1}$. Recall that $gT = g'T$ implies that $Tg^{-1} = T(g')^{-1}$. It follows that

$$gTg^{-1} = (g'T)g^{-1} = g'(Tg^{-1}) = g'(T(g')^{-1}) = g'T(g')^{-1}$$

This is a contradiction and so π must be injective. Let $gT \in G/T$ we have that $\pi(gTg^{-1}) = gT$ for any g a representative of gT and so π is surjective. We have thus, found the bijection from $\{gTg^{-1} : g \in G\}$ to G/T. □

Theorem 3.9. *There are no finite simple single layer groups*

Proof. Suppose there exists G a finite simple single layer group and let N be the set of all primes dividing $|G|$. For each $p \in N$ let K_p be the number of elements T in $\mathcal{T}(G)$ such that $|T| = p$. If $|G| = p^k$ for some $k > 2$ then $|G|$ is not simple. We therefore have that each prime in N has multiplicity one in $|G|$. This means that every Sylow subgroup of G has prime order and conversely that every group of prime order is a Sylow subgroup of G. Since given a prime p all Sylow p-subgroups are conjugate,

$$K_{|T|} = \left|\{gTg^{-1} : g \in G\}\right| = |G/T| = |G|/|T|$$

for all $T \in \mathcal{T}(G)$. Notice that $|\mathcal{T}(G)| > 2$ since both $\mathbb{Z}/p^2\mathbb{Z}$ and $\mathbb{Z}/pq\mathbb{Z}$ are not

simple for any two primes p and q. Observe that

$$\begin{aligned}|G| &= 1 + \sum_{p\in N} K_p(p-1) = 1 + \sum_{p\in N}(K_p p - K_p) = 1 + \sum_{p\in N}|G| - \sum_{p\in N}\frac{|G|}{p} \\ &= 1 + |N||G| - |G|\sum_{p\in N}\frac{1}{p}\end{aligned}$$

where we use the fact that

$$|G| = 1 + \sum_{T\in\mathcal{T}(G)}(|T|-1) = 1 + \sum_{p\in N} K_p(p-1)$$

If $|N| - 1 = \sum_{p\in N}\frac{1}{p}$ then

$$-1 = |N||G| - |G| - |G|\sum_{p\in N}\frac{1}{p} = |G|\left(|N| - 1 - \sum_{p\in N}\frac{1}{p}\right) = 0$$

which is a contradiction. Therefore, we may conclude that

$$|G| = \frac{-1}{|N| - 1 - \sum_{p\in N}\frac{1}{p}} = \frac{1}{\sum_{p\in N}\frac{1}{p} + 1 - |N|}$$

In order for G to be a nontrivial group $|G|$ must be a positive integer which is greater than one. If $\sum_{p\in N}\frac{1}{p} + 1 - |N|$ is less than zero then $|G|$ is a negative number and if $\sum_{p\in N}\frac{1}{p} + 1 - |N|$ is greater than or equal to 1 then $|G| \leq 1$, both of which are not possible given G is a nontrivial group. We then have that

$$0 < \sum_{p\in N}\frac{1}{p} + 1 - |N| < 1$$

(the inner term clearly cannot be zero as then $|G|$ would be undefined). This implies that

$$|N| - 1 < \sum_{p\in N}\frac{1}{p} < |N|$$

Let q be the smallest prime in N. We then have that for all $p \in N$, $\frac{1}{p} \leq \frac{1}{q}$ and so

$$\sum_{p \in N} \frac{1}{p} \leq \sum_{p \in N} \frac{1}{q} = \frac{|N|}{q}$$

We then have that

$$|N| - 1 < \frac{|N|}{q}$$

and so

$$\begin{aligned} q|N| - q &< |N| \\ (q-1)|N| &< q \\ |N| &< \frac{q}{q-1} \end{aligned}$$

Since G is not a group of prime order, there must be at least two primes which divide $|G|$. This means that $|N| \geq 2$. We then have that

$$\begin{aligned} 2(q-1) &< q \\ 2q - 2 &< q \\ q &< 2 \end{aligned}$$

Since 2 is the smallest prime, this is a contradiction. Therefore, no finite simple single layer groups may exist. □

4 Conclusion

We conclude that since there are no finite and simple single layer groups, every single layer group must be of the form $\mathbb{Z}/p^2\mathbb{Z}$ or $\mathbb{Z}/p\mathbb{Z} \ltimes \mathbb{Z}/q\mathbb{Z}$ as stated in Theorem 3.4. This means that all finite groups with only maximal nontrivial subgroups have the form of the last sentence.